

Recon-ng Activity

OVERVIEW

The **Recon-ng** application provides a framework that allows penetration testers to quickly perform common open source reconnaissance tasks.

HELPFUL LINKS

- Getting Started Wiki: <https://github.com/lanmaster53/recon-ng/wiki/Getting-Started>
- Tutorial: <https://hackertarget.com/recon-ng-tutorial>
- Cheat Sheet: <https://www.blackhillsinfosec.com/wp-content/uploads/2019/11/recon-ng-5.x-cheat-sheet-Sheet1-1.pdf>

STARTING RECON-NG

From the Kali Linux start menu, select **01 – Information Gathering**. Next, click the icon for the **recon-ng** application.

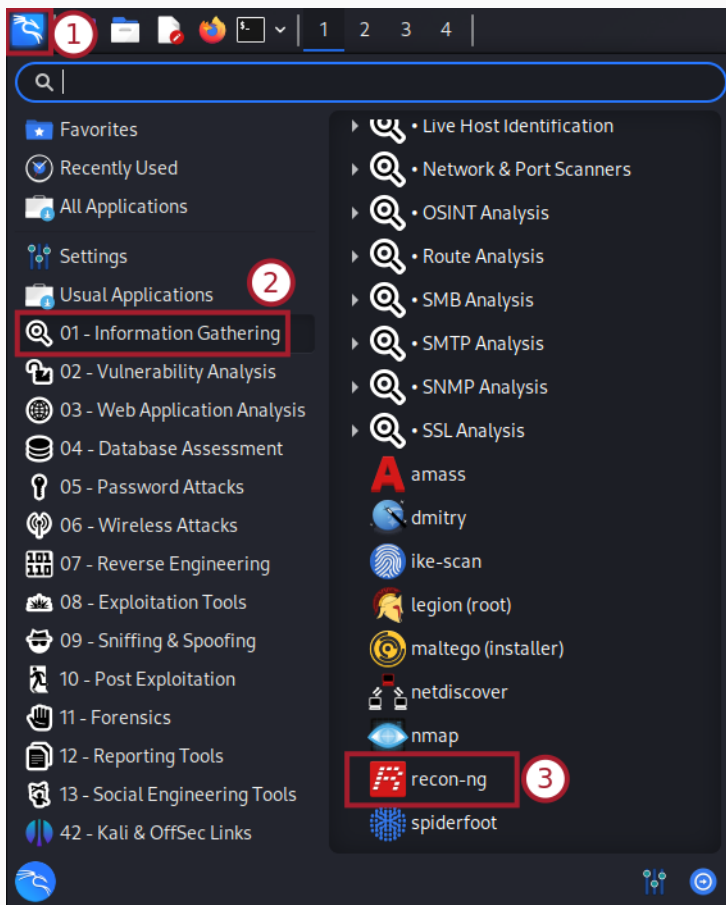


Image 1: Starting recon-ng

The **Recon-ng** command line will display.

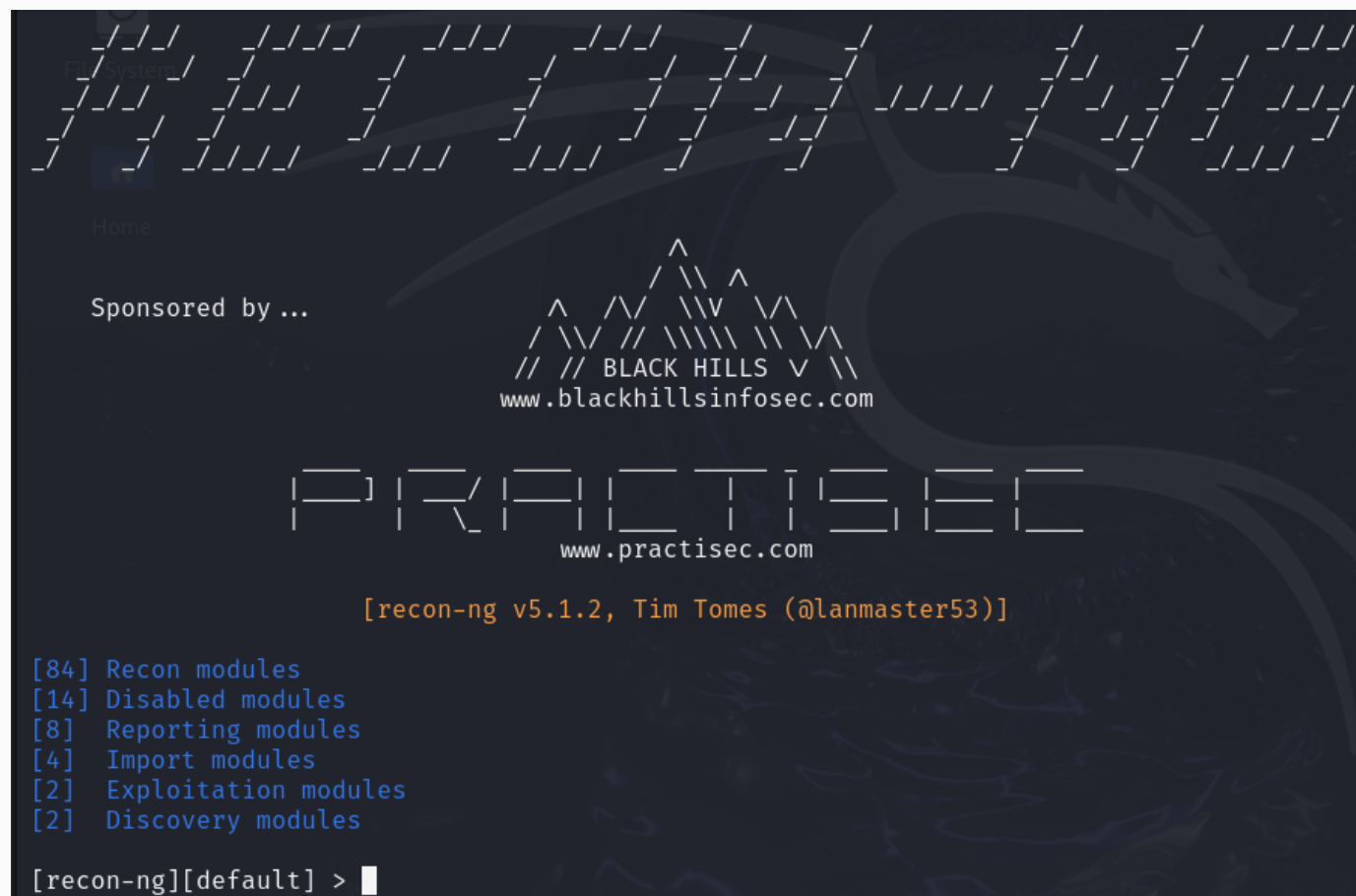


Image 2: recon-ng Command Line

INITIAL RECON-NG STEPS

The **Recon-ng** application makes use of modules to locate open source intelligence items. These modules examine a specific source of information and report what they find. Initially, no modules are installed.

To install all **Recon-ng** modules, run the following command.

```
marketplace install all
```

You may see a number of entries at the end, indicating API keys are not set for some of the modules. These messages may be ignored. To properly work, some modules require such keys.

```
[!] 'github_api' key not set. github_repos module will likely fail at runtime. See 'keys add'.
[!] 'whoxy api' key not set. whoxy_dns module will likely fail at runtime. See 'keys add'.
```

Image 3: Missing API Keys

View the installed modules by entering the **modules search** command.

```
[recon-ng][default] > modules search

Discovery
-----
discovery/info_disclosure/cache_snoop
discovery/info_disclosure/interesting_files

Exploitation
-----
exploitation/injection/command_injector
exploitation/injection/xpath_bruter
```

Image 4: modules search

Next, create a workspace to save all results of your reconnaissance. For this activity, the paloaltonetworks.com domain will be examined.

```
workspaces create PANPenTest
```

Inside your workspace, identify the domain to be used by **Recon-ng** modules.

```
db insert domains
paloaltonetworks.com
```

show domains identifies the domains used in the upcoming module examples.

```
[recon-ng][PANPenTest] > db insert domains
domain (TEXT): paloaltonetworks.com
notes (TEXT): Palo Alto Networks domain
[*] 1 rows affected.
[recon-ng][PANPenTest] > show domains
```

rowid	domain	notes	module
1	paloaltonetworks.com	Palo Alto Networks domain	user_defined

```
[*] 1 rows returned
```

Image 5: Domains

USING RECON-NG MODULES

THE WHOIS_POCS MODULE

This activity starts reconnaissance work by attempting to identify any points of contact for a specified domain. Whois information is a good place to start.

Search for whois-related modules using the following command.

```
modules search whois
```

```
[recon-ng][PANPenTest] > modules search whois
[*] Searching installed modules for 'whois' ...

Recon
-----
recon/companies-domains/viewdns_reverse_whois
recon/companies-multi/whois_miner
recon/domains-companies/whoxy_whois
recon/domains-contacts/whois_pocs
recon/netblocks-companies/whois_orgs
```

Image 6: modules search whois

Next, load the **whois_pocs** module, preparing **Recon-ng** to use it.

```
modules load recon/domains-contacts/whois_pocs
```

Use the **info** command to display information about the module, including required parameters.

```
[recon-ng][PANPenTest][whois_pocs] > info

Name: Whois POC Harvester
Author: Tim Tomes (@lanmaster53)
Version: 1.0

Description:
  Uses the ARIN Whois RWS to harvest POC data from whois queries for the given domain. Updates the
  'contacts' table with the results.

Options:
  Name      Current Value  Required  Description
  -----
  SOURCE    default          yes       source of input (see 'info' for details)

Source Options:
  default   SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
  <string>  string representing a single input
  <path>    path to a file containing a list of inputs
  query <sql> database query returning one column of inputs
```

Image 7: whois_pocs Module Information

For the **whois_pocs** module, a SOURCE domain must be specified. Since you previously provided a domain for your workspace, no additional information is needed.

Use the **run** command to execute the module, causing it to retrieve any related information it can find. If successful, one or more contacts will be identified and added to the workspace.

```
[*] _____
[*] URL: http://whois.arin.net/rest/poc/EXPAN-ARIN
[*] Country: United States
[*] Email: scaninfo@paloaltonetworks.com
[*] First_Name: None
[*] Last_Name: EXPANSE SCANNING
[*] Middle_Name: None
[*] Notes: None
[*] Phone: None
[*] Region: Santa Clara, CA
[*] Title: Whois contact
[*] _____

SUMMARY

[*] 8 total (0 new) contacts found.
```

Image 8: whois_pocs run Results

To see which information has been gathered, use the **dashboard** command.

+-----+	
Results Summary	
+-----+	
Category	Quantity
+-----+	
Domains	1
Companies	0
Netblocks	0
Locations	0
Vulnerabilities	0
Ports	0
Hosts	0
Contacts	5
Credentials	0
Leaks	0
Pushpins	0
Profiles	0
Repositories	0
+-----+	

Image 9: dashboard

Here, some contacts, have been saved to the database.

To view identified contacts, use the **show contacts** command.

rowid	first_name	middle_name	last_name	email	title	region	country	phone
1	Ali		Alshaea	aalshaea@paloaltonetworks.com	Whois contact	Fort Worth, TX	United States	
2			Global Network Services	gns-prod@paloaltonetworks.com	Whois contact	Santa Clara, CA	United States	
3			IT Department	itadmins@paloaltonetworks.com	Whois contact	Santa Clara, CA	United States	
4			IT Department	iaquino@paloaltonetworks.com	Whois contact	Santa Clara, CA	United States	
5			EXPANSE SCANNING	scaninfo@paloaltonetworks.com	Whois contact	Santa Clara, CA	United States	

Image 10: show contacts

THE HACKERTARGET MODULE

The next task in this activity will attempt to identify any known hosts for the domain. The **hackertarget** module provides a useful component for that purpose.

Search for hackertarget-related modules using the following command.

```
modules search hackertarget
```

```
[recon-ng][PANPenTest] > modules search hackertarget
[*] Searching installed modules for 'hackertarget' ...

Recon
-----
recon/domains-hosts/hackertarget
```

Image 11: modules search hackertarget

Next, load the **hackertarget** module.

```
modules load recon/domains-hosts/hackertarget
```

Use the **info** command to display information about the module, including required parameters.

```
[recon-ng][PANPenTest][hackertarget] > info

Name: HackerTarget Lookup
Author: Michael Henriksen (@michenriksen)
Version: 1.1

Description:
  Uses the HackerTarget.com API to find host names. Updates the 'hosts' table with the results.

Options:
  Name      Current Value  Required  Description
  -----
  SOURCE    default             yes       source of input (see 'info' for details)

Source Options:
  default      SELECT DISTINCT domain FROM domains WHERE domain IS NOT NULL
  <string>     string representing a single input
  <path>       path to a file containing a list of inputs
  query <sql>  database query returning one column of inputs
```

Image 12: hackertarget Module Information

For the **hackertarget** module, a SOURCE domain must be specified. Since you previously provided a domain for your workspace, no additional information is needed.

Use the **run** command to execute the module, causing it to retrieve any related information it can find. If successful, one or more new hosts will be identified and added to the workspace.

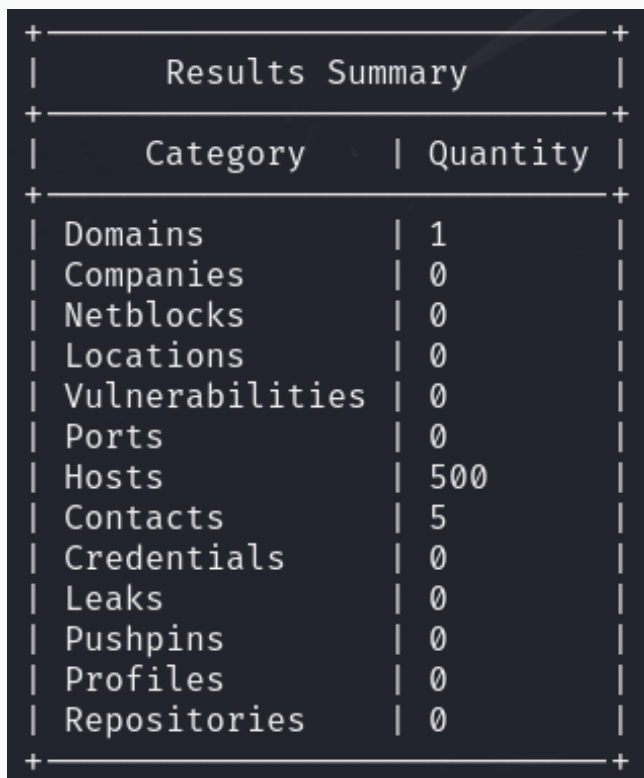
```
[*] -----
[*] Country: None
[*] Host: postg.paloaltonetworks.com
[*] Ip_Address: 199.167.52.59
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*] -----

SUMMARY

[*] 500 total (500 new) hosts found.
```

Image 13: hackertarget run Results

To see which information has been gathered, use the **dashboard** command.

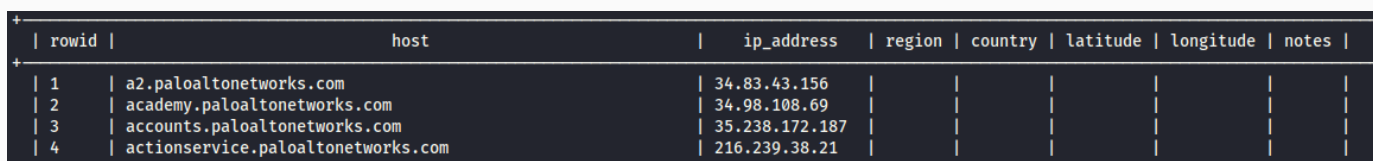


Results Summary	
Category	Quantity
Domains	1
Companies	0
Netblocks	0
Locations	0
Vulnerabilities	0
Ports	0
Hosts	500
Contacts	5
Credentials	0
Leaks	0
Pushpins	0
Profiles	0
Repositories	0

Image 14: dashboard

The identified hosts have been saved to the database.

To view the hosts, use the **show hosts** command.



rowid	host	ip_address	region	country	latitude	longitude	notes
1	a2.paloaltonetworks.com	34.83.43.156					
2	academy.paloaltonetworks.com	34.98.108.69					
3	accounts.paloaltonetworks.com	35.238.172.187					
4	actionservice.paloaltonetworks.com	216.239.38.21					

Image 15: show hosts

RECON-NG REPORTING

Recon-ng provides a number of reporting options, including HTML pages.

Reporting

```
reporting/csv
reporting/html
reporting/json
reporting/list
reporting/proxifier
reporting/pushpin
reporting/xlsx
reporting/xml
```

Image 16: Reporting Modules

Next, load the **html** module.

```
modules load reporting/html
```

Use the **info** command to display information about the module.

```
[recon-ng][PANPenTest][html] > info

Name: HTML Report Generator
Author: Tim Tomes (@lanmaster53)
Version: 1.0

Description:
Creates an HTML report.

Options:

```

Name	Current Value	Required	Description
CREATOR		yes	use creator name in the report footer
CUSTOMER		yes	use customer name in the report header
FILENAME	/home/kali/.recon-ng/workspaces/PANPenTest/results.html	yes	path and filename for report output
SANITIZE	True	yes	mask sensitive data in the report

Image 17: html Module Information

For the **html** module, a CREATOR and CUSTOMER must be specified. Enter your name for the CREATOR and Palo Alto Networks for the CUSTOMER for this activity.

```
options set CREATOR yourname
options set CUSTOMER customername
```

```
[recon-ng][PANPenTest][html] > options set CREATOR student
CREATOR ⇒ student
[recon-ng][PANPenTest][html] > options set CUSTOMER Palo Alto Networks
CUSTOMER ⇒ Palo Alto Networks
[recon-ng][PANPenTest][html] > info

Name: HTML Report Generator
Author: Tim Tomes (@lanmaster53)
Version: 1.0

Description:
Creates an HTML report.

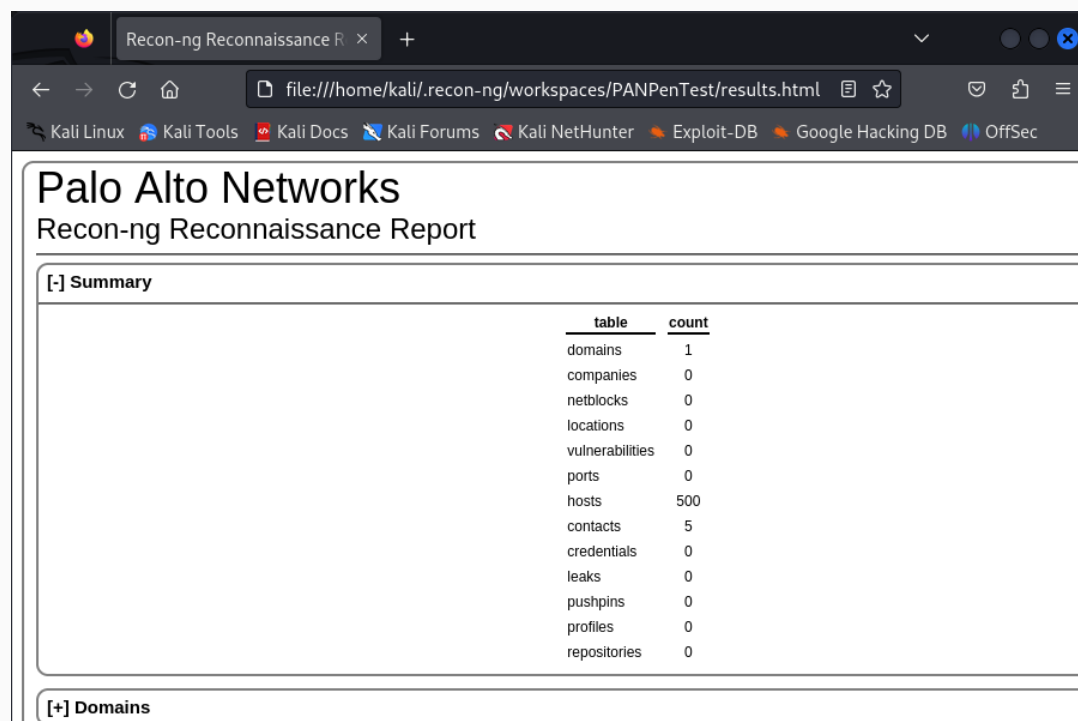
Options:
```

Name	Current Value	Required
CREATOR	student	yes
CUSTOMER	Palo Alto Networks	yes
FILENAME	/home/kali/.recon-ng/workspaces/PANPenTest/results.html	yes
SANITIZE	True	yes

Image 18: Setting Module Options

Use the **run** command to execute the module, causing it to build the HTML report.

Use a web browser to open the file. The FILENAME option identifies the saved report's location.



Palo Alto Networks
Recon-ng Reconnaissance Report

[-] Summary

table	count
domains	1
companies	0
netblocks	0
locations	0
vulnerabilities	0
ports	0
hosts	500
contacts	5
credentials	0
leaks	0
pushpins	0
profiles	0
repositories	0

[+] Domains

Image 19: recon-ng html Report

THE RECON-NG WEB INTERFACE

In addition to the command line, the **Recon-ng** application provides a web interface to view dashboard results.

From a regular command prompt, enter the **recon-web** command.

```
(kali@kali)-[/usr/share/recon-ng]
$ recon-web
*****
* Welcome to Recon-web, the analytics and reporting engine for Recon-ng!
* This is a web-based user interface. Open the URL below in your browser to begin.
* Recon-web includes the Recon-API, which can be accessed via the `/api/` URL.
*****
[*] Marketplace disabled.
[!] 'github_api' key not set. github_repos module will likely fail at runtime. See 'keys add'.
[!] 'whoxy_api' key not set. whoxy_dns module will likely fail at runtime. See 'keys add'.
```

Image 20: recon-web

The end of the output provides the URL used to access the web interface.

```
[*] Version check disabled.
* Workspace initialized: default
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
Press CTRL+C to quit
```

Image 21: http://127.0.0.1:5000

Use a web browser to open the <http://127.0.0.1:5000> site.

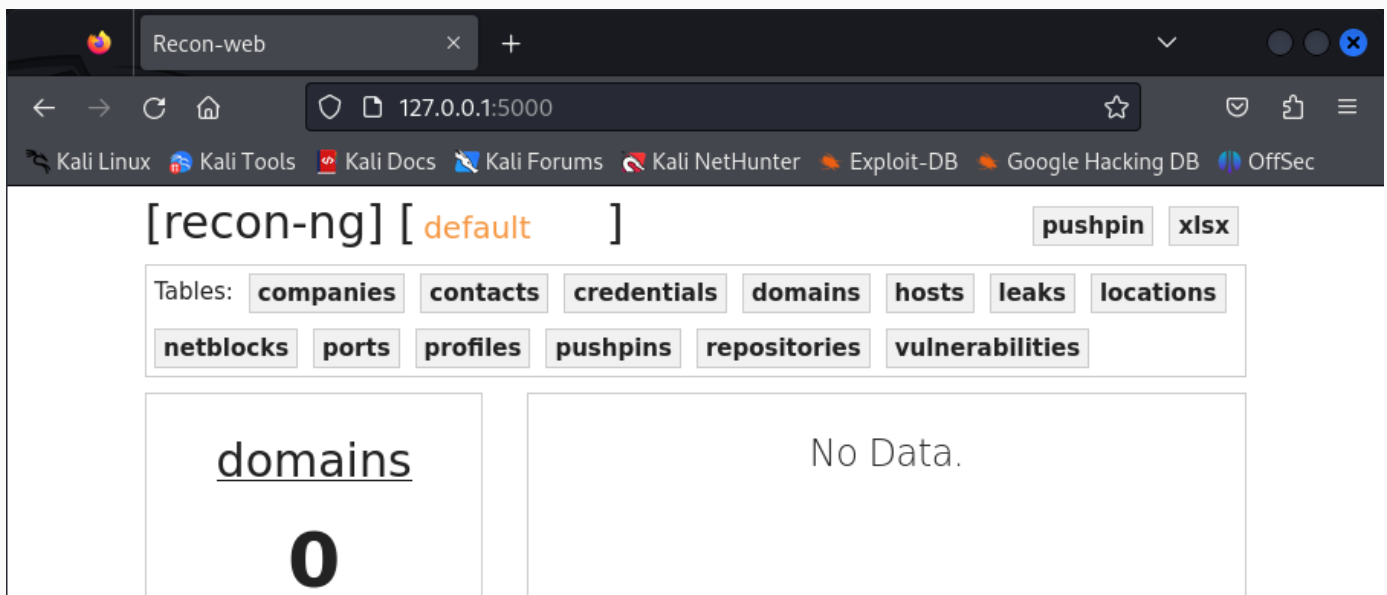


Image 22: recon-web Site

Click the default workspace, listed in orange, to change it

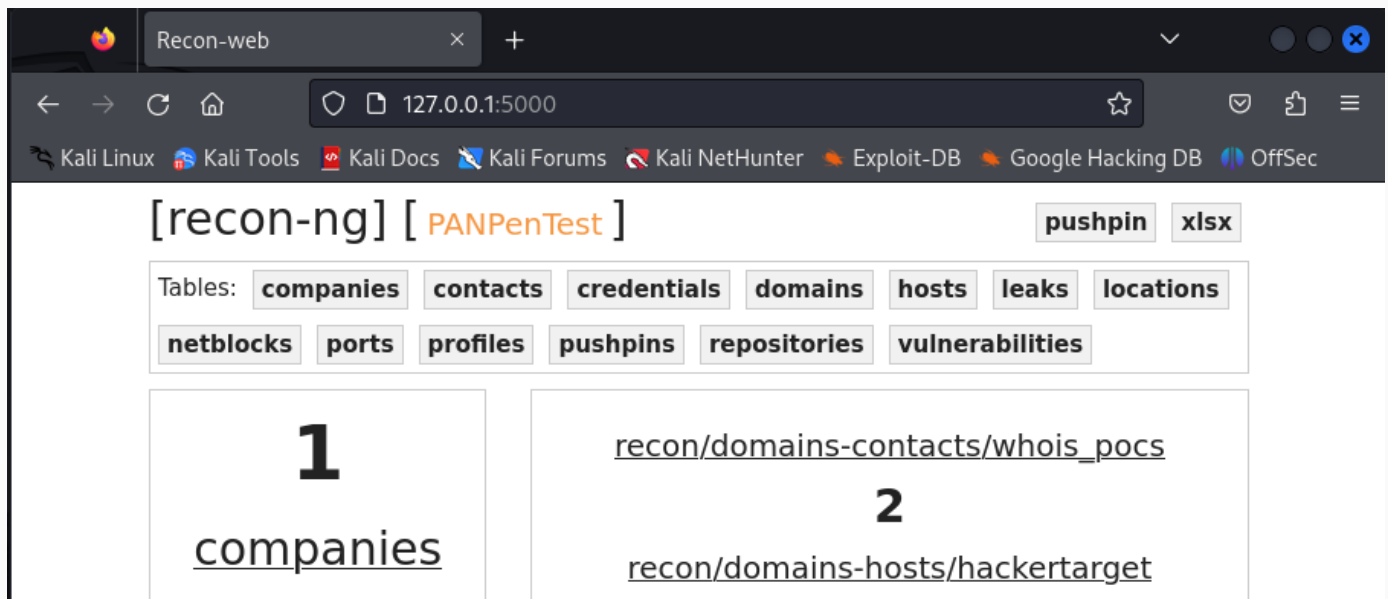


Image 23: recon-web PANPenTest Workspace

You may examine any of the tables, populated by modules that already ran and collected information.

When finished, close your browser, return to the regular command prompt, and press **Ctrl+c** to stop the **recon-web** application.